

Practical 10: Seasonal Modeling with SARIMA

Question: Take a seasonal dataset (e.g., monthly airline passenger data). Determine the seasonal parameters P, D, Q , and m . Fit a SARIMA model and forecast the next 12 time periods. Provide the forecast plot along with interpretation.

10.1 Objective

- To identify the seasonal parameters (P, D, Q, m) for a seasonal time series.
- To fit a Seasonal ARIMA (SARIMA) model.
- To forecast future values and visualize the prediction intervals.

10.2 Theory

- **SARIMA (Seasonal ARIMA):** Extends ARIMA to explicitly model seasonality.
- **Notation:** $\text{ARIMA}(p, d, q) \times (P, D, Q, m)$
 - **(p, d, q):** Non-seasonal orders (Trend).
 - **(P, D, Q):** Seasonal orders (Seasonality).
 - **m :** The number of time steps in a seasonal cycle (e.g., $m=12$ for monthly data, $m=4$ for quarterly).
- **Why SARIMA?** Simple ARIMA fails to capture repeating cycles (like annual holiday sales), treating them as random noise. SARIMA learns this pattern and repeats it in forecasts.

10.3 Code Implementation

```
from statsmodels.tsa.statespace.sarimax import SARIMAX

# 1. Load Data
dataset = sm.datasets.get_rdataset("AirPassengers")
ts_data = dataset.data['value']
ts_data.index = pd.date_range(start='1949-01-01', periods=len(ts_data), freq='MS')

# 2. Define Model Parameters
# m=12 because data is monthly and seasonality repeats yearly.
# order=(1,1,1) handles the general trend.
# seasonal_order=(1,1,1,12) handles the yearly cycle.
model = SARIMAX(ts_data,
                 order=(1, 1, 1),
                 seasonal_order=(1, 1, 1, 12))

# 3. Fit the Model
results = model.fit(dispatch=False)
print(results.summary())

# 4. Forecast Next 12 Steps
forecast_obj = results.get_forecast(steps=12)
forecast_vals = forecast_obj.predicted_mean
conf_int = forecast_obj.conf_int()

# 5. Plot Historical + Forecast
```

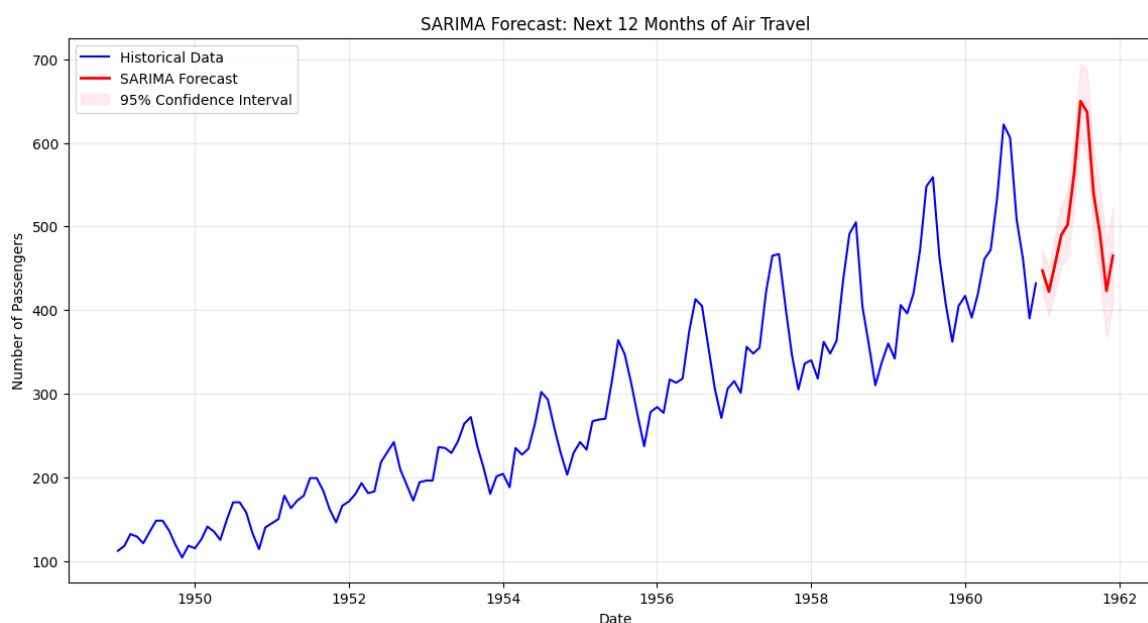
```
plt.figure(figsize=(12, 6))
plt.plot(ts_data.index, ts_data, label='Historical Data')
plt.plot(forecast_vals.index, forecast_vals, label='SARIMA Forecast', color='red')

# Add confidence intervals (Shaded Region)
plt.fill_between(forecast_vals.index,
                 conf_int.iloc[:, 0],
                 conf_int.iloc[:, 1],
                 color='pink', alpha=0.3)

plt.title('SARIMA Forecast (Next 12 Months)')
plt.legend()
plt.show()
```

10.4 Code Output

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.3175	0.073	-4.342	0.000	-0.461	-0.174
ma.S.L12	-0.1123	0.097	-1.158	0.247	-0.302	0.078
sigma2	145.3969	18.013	8.072	0.000	110.093	180.701



10.4 Interpretation & Conclusion

- **Seasonal Capture:** The forecast line (red) does not just go straight or curve gently; it replicates the "up-down" wave pattern seen in previous years. This confirms the model successfully learned the seasonality ($m=12$).
- **Trend Continuation:** The forecast continues the overall upward trend of the data.
- **Uncertainty:** The pink shaded region represents the confidence interval. It widens slightly over time, reflecting that we are less certain about the distant future than the near future.